

Penalty and gradient (Pen-and-grad) for Envi-Flux

March 13, 2024

1 The cost function

The cost function is:

$$J(\chi) = \underbrace{\frac{1}{2}\chi^\top\chi}_{J_b} + \underbrace{\frac{1}{2}\sum_{t=1}^T (\mathbf{y}_t - \mathbf{H}_t\mathbf{M}_{0\rightarrow t}\mathbf{C}[\mathbf{U}\chi + \mathbf{x}_b])^\top \mathbf{R}_t^{-1} (\mathbf{y}_t - \mathbf{H}_t\mathbf{M}_{0\rightarrow t}\mathbf{C}[\mathbf{U}\chi + \mathbf{x}_b])}_{J_o}. \quad (1)$$

The time t is measured in the model's own major timestep units (Δt , e.g. 12 hours). Given a time t , the observations $\mathbf{y}_t$ correspond to actual times τ , $t-1 \leq \tau < t$. χ is the control variable, and $\mathbf{U}$ is the control variable transform. $\mathbf{U}\chi$ returns an augmented state comprising the initial state and that source field at set of times:

$$\mathbf{x} = \mathbf{U}\chi = \begin{pmatrix} \mathbf{x}_0 \\ \boldsymbol{\rho} \end{pmatrix}, \quad (2)$$

where there are γ steps per flux timestep ($\gamma \in \mathbb{N}^+$) and there are $n_\gamma + 1$ flux fields:

$$\boldsymbol{\rho} = \begin{pmatrix} \rho_0 \\ \rho_\gamma \\ \vdots \\ \rho_{n_\gamma\gamma} \end{pmatrix}. \quad (3)$$

The observation operator, $\mathbf{H}_t$, needs to know the state at times $t-1$ and t (to predict the observations in $\mathbf{y}_t$ above). Hence introduce the state labeled by a capital $\mathbf{X}_t$, which

contains state pairs ($\mathbf{x}_{t-1}$ and $\mathbf{x}_t$), and the fluxes:

$$\mathbf{X}_t = \begin{pmatrix} \mathbf{x}_{t-1} \\ \mathbf{x}_t \\ \boldsymbol{\rho} \end{pmatrix}, \quad (4)$$

with

$$\mathbf{X}_0 = \begin{pmatrix} \mathbf{0} \\ \mathbf{x}_0 \\ \boldsymbol{\rho} \end{pmatrix}. \quad (5)$$

This $\mathbf{X}_0$ is different from $\mathbf{x}$ in (2), but $\mathbf{X}_0$ can be constructed from $\mathbf{x}$ as follows:

$$\mathbf{X}_0 = \mathbf{C}\mathbf{x} = \begin{pmatrix} \mathbf{0} & \mathbf{0} \\ \mathbf{I} & \mathbf{0} \\ \mathbf{0} & \mathbf{I} \end{pmatrix} \mathbf{x}. \quad (6)$$

The observation operator acts as follows:

$$\mathbf{H}_t\mathbf{X}_t = \mathbf{H}_t \begin{pmatrix} \mathbf{x}_{t-1} \\ \mathbf{x}_t \\ \boldsymbol{\rho} \end{pmatrix}. \quad (7)$$

The time-stepping of the state vector is defined as

$$\mathbf{X}_t = \mathbf{M}_{0\rightarrow t}\mathbf{X}_0 = \mathbf{M}_{t-1\rightarrow t} \cdots \mathbf{M}_{1\rightarrow 2}\mathbf{M}_{0\rightarrow 1}\mathbf{X}_0. \quad (8)$$

This (calligraphic $\mathbf{M}_{t\rightarrow t+1}$) propagates the tracer pairs, but keeps the fluxes in the state vector. In terms of the propagation of the tracer $\mathbf{x}_t$ to $\mathbf{x}_{t+1}$, the time-stepping is (Roman $\mathbf{M}_{t\rightarrow t+1}$)

$$\delta\mathbf{x}_{t+1} = \mathbf{M}_{t\rightarrow t+1}\delta\mathbf{x}_t + \mathbf{S}\boldsymbol{\rho}^{I([t+1]/\gamma)}\Delta t. \quad (9)$$

Here the function $I(\bullet)$ integerises its argument, so time t is associated with flux time $I(t/\gamma)$. Re-indexing (9) gives

$$\delta\mathbf{x}_t = \mathbf{M}_{t-1\rightarrow t}\delta\mathbf{x}_{t-1} + \mathbf{S}\boldsymbol{\rho}^{I(t/\gamma)}\Delta t. \quad (10)$$

Putting this together gives the relationship between $\mathbf{M}_{t-1\rightarrow t}$ and $\mathbf{M}_{t-1\rightarrow t}$:

or

$$\begin{aligned} \mathbf{X}_t &= \mathcal{M}_{t-1 \rightarrow t} \mathbf{X}_{t-1} \\ \begin{pmatrix} \mathbf{x}_{t-1} \\ \mathbf{x}_t \\ \boldsymbol{\rho} \end{pmatrix} &= \begin{pmatrix} \mathbf{0} & \mathbf{I} & \mathbf{0} \\ \mathbf{0} & \mathbf{M}_{t-1 \rightarrow t} & \mathbf{S} \\ \mathbf{0} & \mathbf{0} & \mathbf{I} \end{pmatrix} \begin{pmatrix} \mathbf{x}_{t-2} \\ \mathbf{x}_{t-1} \\ \boldsymbol{\rho} \end{pmatrix} \end{aligned} \quad (11)$$

$$\begin{aligned} \mathbf{X}_{t+1} &= \mathcal{M}_{t \rightarrow t+1} \mathbf{X}_t \\ \begin{pmatrix} \mathbf{x}_t \\ \mathbf{x}_{t+1} \\ \boldsymbol{\rho} \end{pmatrix} &= \begin{pmatrix} \mathbf{0} & \mathbf{I} & \mathbf{0} \\ \mathbf{0} & \mathbf{M}_{t \rightarrow t+1} & \mathbf{S} \\ \mathbf{0} & \mathbf{0} & \mathbf{I} \end{pmatrix} \begin{pmatrix} \mathbf{x}_{t-1} \\ \mathbf{x}_t \\ \boldsymbol{\rho} \end{pmatrix}. \end{aligned} \quad (12)$$

The adjoint of this step is

or

$$\begin{aligned} \hat{\mathbf{X}}_{t-1} &= \mathcal{M}_{t-1 \rightarrow t}^\top \hat{\mathbf{X}}_t \\ \begin{pmatrix} \hat{\mathbf{x}}_{t-2} \\ \hat{\mathbf{x}}_{t-1} \\ \hat{\boldsymbol{\rho}} \end{pmatrix} &= \begin{pmatrix} \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{I} & \mathbf{M}_{t-1 \rightarrow t}^\top & \mathbf{0} \\ \mathbf{0} & \mathbf{S}^\top & \mathbf{I} \end{pmatrix} \begin{pmatrix} \hat{\mathbf{x}}_{t-1} \\ \hat{\mathbf{x}}_t \\ \hat{\boldsymbol{\rho}} \end{pmatrix} \end{aligned} \quad (13)$$

$$\begin{aligned} \hat{\mathbf{X}}_t &= \mathcal{M}_{t \rightarrow t+1}^\top \hat{\mathbf{X}}_{t+1} \\ \begin{pmatrix} \hat{\mathbf{x}}_{t-1} \\ \hat{\mathbf{x}}_t \\ \hat{\boldsymbol{\rho}} \end{pmatrix} &= \begin{pmatrix} \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{I} & \mathbf{M}_{t \rightarrow t+1}^\top & \mathbf{0} \\ \mathbf{0} & \mathbf{S}^\top & \mathbf{I} \end{pmatrix} \begin{pmatrix} \hat{\mathbf{x}}_t \\ \hat{\mathbf{x}}_{t+1} \\ \hat{\boldsymbol{\rho}} \end{pmatrix}. \end{aligned} \quad (14)$$

2 The gradient of the cost function

The gradient of J_b is trivial:

$$\frac{\partial J_b}{\partial \boldsymbol{\chi}} = \boldsymbol{\chi}. \quad (15)$$

The gradient of J_o is non-trivial and is found as follows.

$$\begin{aligned} \frac{\partial J_o}{\partial \boldsymbol{\chi}} &= \frac{\partial}{\partial \boldsymbol{\chi}} \left[\frac{1}{2} \sum_{t=1}^T (\mathbf{y}_t - \mathbf{H}_t \mathcal{M}_{0 \rightarrow t} \mathbf{C} [\mathbf{U} \boldsymbol{\chi} + \mathbf{x}_b])^\top \mathbf{R}_t^{-1} (\mathbf{y}_t - \mathbf{H}_t \mathcal{M}_{0 \rightarrow t} \mathbf{C} [\mathbf{U} \boldsymbol{\chi} + \mathbf{x}_b]) \right] \\ &= -\mathbf{U}^\top \mathbf{C}^\top \sum_{t=1}^T \mathcal{M}_{0 \rightarrow t}^\top \mathbf{H}_t^\top \mathbf{R}_t^{-1} (\mathbf{y}_t - \mathbf{H}_t \mathcal{M}_{0 \rightarrow t} \mathbf{C} [\mathbf{U} \boldsymbol{\chi} + \mathbf{x}_b]) \\ &= \mathbf{U}^\top \mathbf{C}^\top \sum_{t=1}^T \mathcal{M}_{0 \rightarrow t}^\top \mathbf{H}_t^\top \frac{\partial J_o}{\partial \mathbf{y}_t^m}, \end{aligned} \quad (16)$$

where $\partial J_o / \partial \mathbf{y}_t^m = -\mathbf{R}_t^{-1} (\mathbf{y}_t - \mathbf{y}_t^m)$ and $\mathbf{y}_t^m = \mathbf{H}_t \mathcal{M}_{0 \rightarrow t} \mathbf{C} [\mathbf{U} \boldsymbol{\chi} + \mathbf{x}_b]$.

$$\begin{aligned} \frac{\partial J_o}{\partial \boldsymbol{\chi}} &= \mathbf{U}^\top \mathbf{C}^\top \begin{bmatrix} \mathcal{M}_{0 \rightarrow 1}^\top & \mathcal{M}_{1 \rightarrow 2}^\top & \cdots & \mathcal{M}_{t-1 \rightarrow t}^\top & \cdots & \mathcal{M}_{T-2 \rightarrow T-1}^\top & \mathcal{M}_{T-2 \rightarrow T-1}^\top & \mathcal{M}_{T-1 \rightarrow T}^\top & \mathbf{H}_T^\top \frac{\partial J_o}{\partial \mathbf{y}_T^m} \\ \mathcal{M}_{0 \rightarrow 1}^\top & \mathcal{M}_{1 \rightarrow 2}^\top & \cdots & \mathcal{M}_{t-1 \rightarrow t}^\top & \cdots & \mathcal{M}_{T-2 \rightarrow T-1}^\top & \mathcal{M}_{T-2 \rightarrow T-1}^\top & \mathbf{H}_{T-1}^\top \frac{\partial J_o}{\partial \mathbf{y}_{T-1}^m} + \\ \mathcal{M}_{0 \rightarrow 1}^\top & \mathcal{M}_{1 \rightarrow 2}^\top & \cdots & \mathcal{M}_{t-1 \rightarrow t}^\top & \cdots & \mathcal{M}_{T-2 \rightarrow T-1}^\top & \mathbf{H}_{T-2}^\top \frac{\partial J_o}{\partial \mathbf{y}_{T-2}^m} + \\ \vdots & \vdots & \ddots & \vdots & & & & & \\ \mathcal{M}_{0 \rightarrow 1}^\top & \mathcal{M}_{1 \rightarrow 2}^\top & \cdots & \mathcal{M}_{t-1 \rightarrow t}^\top & \mathbf{H}_t^\top \frac{\partial J_o}{\partial \mathbf{y}_t^m} & & & & \\ \vdots & \vdots & & & & & & & \\ \mathcal{M}_{0 \rightarrow 1}^\top & \mathcal{M}_{1 \rightarrow 2}^\top & \mathbf{H}_2^\top \frac{\partial J_o}{\partial \mathbf{y}_2^m} + \\ \mathcal{M}_{0 \rightarrow 1}^\top & \mathbf{H}_1^\top \frac{\partial J_o}{\partial \mathbf{y}_1^m} \end{bmatrix} \end{aligned}$$

This can be represented as one of the following equivalent algorithms.

2.1 Algorithm 1

- Let $\boldsymbol{\lambda}_t = \partial J_o / \partial \mathbf{X}_t = \hat{\mathbf{X}}_t$ (a column vector).
- $\boldsymbol{\lambda}_T = \mathbf{H}_T^\top \frac{\partial J_o}{\partial \mathbf{y}_T^m}$
- $\boldsymbol{\lambda}_{T-1} = \mathcal{M}_{T-1 \rightarrow T}^\top \boldsymbol{\lambda}_T + \mathbf{H}_{T-1}^\top \frac{\partial J_o}{\partial \mathbf{y}_{T-1}^m}$
- $\boldsymbol{\lambda}_{T-2} = \mathcal{M}_{T-2 \rightarrow T-1}^\top \boldsymbol{\lambda}_{T-1} + \mathbf{H}_{T-2}^\top \frac{\partial J_o}{\partial \mathbf{y}_{T-2}^m}$
- ...
- $\boldsymbol{\lambda}_1 = \mathcal{M}_{1 \rightarrow 2}^\top \boldsymbol{\lambda}_2 + \mathbf{H}_1^\top \frac{\partial J_o}{\partial \mathbf{y}_1^m}$
- $\boldsymbol{\lambda}_0 = \mathcal{M}_{0 \rightarrow 1}^\top \boldsymbol{\lambda}_1$

In general this can be written as the following.

- Let $\boldsymbol{\lambda}_{T+1} = \mathbf{0}$

- Loop from $t = T$ to $t = 1$ backwards.

$$- \quad \lambda_t = \mathcal{M}_{t \rightarrow t+1}^\top \lambda_{t+1} + \mathbf{H}_t^\top \frac{\partial J_o}{\partial \mathbf{y}_t^m}$$

2.2 Algorithm 2

- Let $\lambda_t = \partial J_o / \partial \mathbf{X}_t = \hat{\mathbf{X}}_t$ (a column vector).
- $\lambda_{T-1} = \mathcal{M}_{T-1 \rightarrow T}^\top \mathbf{H}_T^\top \frac{\partial J_o}{\partial \mathbf{y}_T^m}$
- $\lambda_{T-2} = \mathcal{M}_{T-2 \rightarrow T-1}^\top \left(\lambda_{T-1} + \mathbf{H}_{T-1}^\top \frac{\partial J_o}{\partial \mathbf{y}_{T-1}^m} \right)$
- $\lambda_{T-3} = \mathcal{M}_{T-3 \rightarrow T-2}^\top \left(\lambda_{T-2} + \mathbf{H}_{T-2}^\top \frac{\partial J_o}{\partial \mathbf{y}_{T-2}^m} \right)$
- $\dots$
- $\lambda_1 = \mathcal{M}_{1 \rightarrow 2}^\top \left(\lambda_2 + \mathbf{H}_2^\top \frac{\partial J_o}{\partial \mathbf{y}_2^m} \right)$

In terms of the $\hat{\mathbf{X}}_t$, algorithm 2 is

- Let $\lambda_T = \mathbf{0}$
- Loop from $t = T$ to $t = 1$ backwards.

$$- \quad \begin{pmatrix} \hat{\mathbf{x}}_{t-2} \\ \hat{\mathbf{x}}_{t-1} \\ \hat{\rho} \end{pmatrix} = \begin{pmatrix} \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{I} & \mathbf{M}_{t-1 \rightarrow t}^\top & \mathbf{0} \\ \mathbf{0} & \mathbf{S}^\top & \mathbf{I} \end{pmatrix} \begin{pmatrix} \hat{\mathbf{x}}_{t-1} \\ \hat{\mathbf{x}}_t \\ \hat{\rho} \end{pmatrix} + \mathbf{H}_t^\top \frac{\partial J_o}{\partial \mathbf{y}_t^m}$$

- The total gradient is $\nabla_{\boldsymbol{\chi}} J = \boldsymbol{\chi} + \mathbf{U}^\top \mathbf{C}^\top \lambda_0$

$$\bullet \quad \lambda_0 = \mathcal{M}_{0 \rightarrow 1}^\top \lambda_1$$

- The total gradient is $\nabla_{\boldsymbol{\chi}} J = \boldsymbol{\chi} + \mathbf{U}^\top \mathbf{C}^\top \lambda_0$

$$\bullet \quad \lambda_0 = \mathcal{M}_{0 \rightarrow 1}^\top \left(\lambda_1 + \mathbf{H}_1^\top \frac{\partial J_o}{\partial \mathbf{y}_1^m} \right)$$

In general this can be written as the following.

- Let $\lambda_T = \mathbf{0}$
- Loop from $t = T$ to $t = 1$ backwards.
- $$- \quad \lambda_{t-1} = \mathcal{M}_{t-1 \rightarrow t}^\top \left(\lambda_t + \mathbf{H}_t^\top \frac{\partial J_o}{\partial \mathbf{y}_t^m} \right)$$
- The total gradient is $\nabla_{\boldsymbol{\chi}} J = \boldsymbol{\chi} + \mathbf{U}^\top \mathbf{C}^\top \lambda_0$

3 Testing the gradient

The gradient test is done by considering the following cost function expanded to first order:

$$J(\boldsymbol{\chi}_0 + \alpha \delta \boldsymbol{\chi}) = J(\boldsymbol{\chi}_0) + \alpha \delta \boldsymbol{\chi}^\top \nabla_{\boldsymbol{\chi}} J + O(\alpha^2).$$

Rearrange this:

$$\frac{J(\boldsymbol{\chi}_0 + \alpha \delta \boldsymbol{\chi}) - J(\boldsymbol{\chi}_0)}{\alpha \delta \boldsymbol{\chi}^\top \nabla_{\boldsymbol{\chi}} J} = 1 + O(\alpha) \equiv \Phi(\alpha)$$